

MOAAZ MOHAMED MAHMOUD ELGHAMRY

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PROFESSIONAL SUMMARY

High-caliber Mechatronics Engineering student with a near-perfect **3.96/4.0 CGPA**, recognized for exceptional technical rigor and multidisciplinary innovation. A dual-certified **SOLIDWORKS Professional (CSWP)** and **Associate (CSWA)** with a specialized focus on **Hardware-Software Co-design**. Expert in architecting distributed automation ecosystems utilizing **Django 5.2**, **Asynchronous Task Queues (Celery/Redis)**, and **Real-time WebSockets**. Proven expertise in **Industrial Automation (Siemens S7-1200/1500)** and **Solar PV Systems**, with a unique ability to integrate **Generative AI (LLMs)** into industrial diagnostics. Adept at leveraging **Linux (WSL/Ubuntu)** environments to deploy high-concurrency engineering solutions.

EDUCATION

Bachelor of Mechatronics Engineering 2023 – Present
Zagazig National University, Egypt **Cumulative GPA: 3.96/4.0**

- **Academic Distinction:** Consistently ranked 1st in the department; recipient of the University Excellence Award for three consecutive years.
- **Technical Specialization:** Advanced Robotics & Kinematics, Control Systems Engineering (PID/LQR/H-infinity), Embedded Systems (AVR/ARM), Machine Design, and Finite Element Analysis (FEA).
- **Research Interests:** Autonomous Navigation, Digital Twin Technology, and AI-driven Predictive Maintenance for Mechatronic Systems.

TECHNICAL PROFICIENCIES

- **Mechanical Design & Engineering:** **SOLIDWORKS (CSWP/CSWA)**, DFM/DFA compliance, Stress/Thermal Analysis (FEA), Fluid Simulation (CFD), Proteus 9 Circuit Prototyping, Solar PV System Design (PVSyst/Standardized Yield Analysis).
- **Software Architecture & Backend:** **Django 5.2 (Expert)**, Django Rest Framework (DRF), **Channels (WebSockets)**, **Celery**, **Redis**, Daphne/Uvicorn/Gunicorn deployment, PostgreSQL (psycopg2), Fernet Cryptography.
- **Industrial Automation:** **Advanced PLC Programming (Siemens TIA Portal)**, HMI Design, SCADA integration, Industrial Communication Protocols (Profinet, Modbus TCP/IP), Motor Control (VFDs/Servos).
- **Data Science & AI Integration:** **Large Language Model (LLM) Integration (Groq, Google GenAI)**, **Pandas**, Openpyxl, NumPy, Python Automation (Requests, BeautifulSoup), Matplotlib.
- **DevOps & OS Management:** **Linux for Windows (WSL)**, Ubuntu Server Management, Bash Scripting, Version Control (Git/GitHub), Environment Virtualization (Pipenv/Venv).

PROFESSIONAL EXPERIENCE & TRAINING

Advanced PLC Specialist Training (Siemens S7-1500) Jan 2025
DON-BOSCO Institute, Cairo, Egypt

- Engineered high-level automation logic for synchronized multi-axis production lines, utilizing SCL and STL programming languages.
- Implemented complex PID control loops for temperature and pressure regulation in industrial manufacturing simulations.
- Optimized Human-Machine Interface (HMI) dashboards to provide real-time diagnostic feedback, reducing operator error rates in high-speed environments.

Industrial Maintenance & Diagnostics Intern 2025
ManTrac (Official Caterpillar Dealer), Egypt

- Conducted hardware-level diagnostics on heavy-duty Caterpillar engine control units (ECUs), interpreting CAN-bus data and SIS technical manuals.
- Executed comprehensive preventative maintenance cycles on hydraulic and power-train systems for industrial machinery.

Systems Engineering & Control Intern

2024

HA Consulting Group, Cairo, Egypt

- Developed technical schematics and wiring diagrams for industrial control panels, ensuring 100% compliance with IEC/NEC standards.
- Automated documentation workflows for electrical systems using custom scripts, increasing departmental drafting speed by 30%.

STRATEGIC TECHNICAL PROJECTS

Unified Mechatronics System Platform (UMSP) | znue-portal.me

2025

- **Architectural Design:** Built a high-availability portal using **Django 5.2**, implementing **Redis** as a message broker for **Celery** background workers.
- **Real-time Interaction:** Integrated **Django Channels** and **Daphne** to facilitate full-duplex WebSocket communication for real-time sensor telemetry.
- **AI-Driven Intelligence:** Developed a Retrieval-Augmented Generation (RAG) system utilizing **Groq** and **Google Generative AI** to analyze complex engineering datasheets and provide instant technical insights.
- **Security & Performance:** Employed **django-fernet-fields** for database-level encryption and optimized SQL queriesets to handle thousands of concurrent relational records.

Industrial Data Ecosystem & ERP Integration

2024

- Engineered a centralized hub for managing mechatronic resource allocations using **React** and **Django REST Framework**.
- Automated data ingestion from legacy Excel formats into PostgreSQL using **Pandas** and **Openpyxl**, maintaining 100% data integrity.

CERTIFICATIONS & PROFESSIONAL MEMBERSHIPS

- **Certified SOLIDWORKS Professional (CSWP)** - Mechanical Design | ID: C-URCXPDYST4
- **Certified SOLIDWORKS Associate (CSWA)** - Mechanical Design | ID: C-EBAX9R8E3C
- **Solar Energy Systems Specialist** - **Egyptian Engineering Syndicate** & Engovation Academy
- **Language Proficiency:** **Arabic** (Native); **English** (C1 - Full Professional Proficiency; Academic/Technical fluency).